

Verified Research Papers — Expanded Collection

20 scholarly papers curated around tool selection, tool learning, planning and evaluation

1. How to use this collection

The assignment requires at least 20 verified scholarly papers, categorized meaningfully, with authors, year, venue/identifier, a link, and a one-line relevance explanation. The list below is organized around the research pipeline rather than alphabetical order.

Survey / Review

Tool Learning with Large Language Models: A Survey

Authors: Changle Qu et al. | **Year:** 2024

Why relevant: Systematizes tool learning into task planning, tool selection, tool calling and response generation, and summarizes benchmarks/evaluation.

Official source: <https://arxiv.org/abs/2405.17935>

Foundation

ReAct: Synergizing Reasoning and Acting in Language Models

Authors: Shunyu Yao et al. | **Year:** 2023

Why relevant: Connects reasoning traces with actions in interactive environments, providing a foundation for action/tool use.

Official source: <https://arxiv.org/abs/2210.03629>

Toolformer: Language Models Can Teach Themselves to Use Tools

Authors: Timo Schick et al. | **Year:** 2023

Why relevant: Studies when to call APIs, which APIs to call and how to construct arguments.

Official source: <https://arxiv.org/abs/2302.04761>

Tool learning

ToolLLM: Facilitating Large Language Models to Master 16000+ Real-world APIs

Authors: Yujia Qin et al. | **Year:** 2023

Why relevant: Introduces large-scale API/tool learning with ToolBench and a tool-use framework.

Official source: <https://arxiv.org/abs/2307.16789>

ToolCoder: A Systematic Code-Empowered Tool Learning Framework for Large Language Models

Authors: Hanxing Ding et al. | **Year:** 2025

Why relevant: Uses code-oriented tool learning with planning, execution, reuse and debugging.

Official source: <https://aclanthology.org/2025.acl-long.874/>

Retrieval

AnyTool: Self-Reflective, Hierarchical Agents for Large-Scale API Calls

Authors: Yu Du et al. | **Year:** 2024

Why relevant: Hierarchical API retrieval and self-reflection for large tool collections.

Official source: <https://proceedings.mlr.press/v235/du24h.html>

ToolGen: Unified Tool Retrieval and Calling via Generation

Authors: Renxi Wang et al. | **Year:** 2024

Why relevant: Unifies tool retrieval and calling through generation and evaluates large tool collections.

Official source: <https://arxiv.org/abs/2410.03439>

ToolNet: Connecting Large Language Models with Massive Tools via Tool Graph

Authors: Xukun Liu et al. | **Year:** 2024

Why relevant: Uses a graph structure to connect and navigate large tool libraries.

Official source: <https://arxiv.org/abs/2403.00839>

Planning

ToolPlanner: A Tool Augmented LLM for Multi Granularity Instructions with Path Planning and Feedback

Authors: Qinzhuo Wu et al. | **Year:** 2024

Why relevant: Uses path planning and feedback for multi-step tool-augmented instructions.

Official source: <https://aclanthology.org/2024.emnlp-main.1018/>

Evaluation

The Berkeley Function Calling Leaderboard (BFCL): From Tool Use to Agentic Evaluation of Large Language Models

Authors: Shishir G. Patil et al. | **Year:** 2025

Why relevant: Evaluates function calling with serial/parallel calls and stateful multi-step agentic settings.

Official source: <https://proceedings.mlr.press/v267/patil25a.html>

AgentBench: Evaluating LLMs as Agents

Authors: Xiao Liu et al. | **Year:** 2023

Why relevant: Multi-environment benchmark for assessing LLMs as agents.

Official source: <https://arxiv.org/abs/2308.03688>

GAIA: a benchmark for General AI Assistants

Authors: Grégoire Mialon et al. | **Year:** 2024

Why relevant: Tests general AI assistants on real-world questions requiring reasoning and tool use.

Official source: https://proceedings.iclr.cc/paper_files/paper/2024/hash/25ae35b5b1738d80f1f03a8713e405ec-Abstract-Conference.html

T-Eval: Evaluating the Tool Utilization Capability of Large Language Models Step by Step

Authors: Zehui Chen et al. | **Year:** 2024

Why relevant: Decomposes tool utilization into instruction following, planning, reasoning, retrieval, understanding and review.

Official source: <https://aclanthology.org/2024.acl-long.515/>

Planning, Creation, Usage: Benchmarking LLMs for Comprehensive Tool Utilization in Real-World Complex Scenarios

Authors: Shijue Huang et al. | **Year:** 2024

Why relevant: UltraTool examines planning, tool creation and tool usage in complex scenarios.

Official source: <https://aclanthology.org/2024.findings-acl.259/>

StableToolBench: Towards Stable Large-Scale Benchmarking on Tool Learning of Large Language Models

Authors: Zhicheng Guo et al. | **Year:** 2024

Why relevant: Addresses stability and reproducibility in large-scale tool-learning evaluation.

Official source: <https://aclanthology.org/2024.findings-acl.664/>

ToolSandbox: A Stateful, Conversational, Interactive Evaluation Benchmark for LLM Tool Use Capabilities

Authors: Jiarui Lu et al. | **Year:** 2025

Why relevant: Evaluates stateful tool execution, dependencies, conversations and dynamic intermediate/final milestones.

Official source: <https://aclanthology.org/2025.findings-naacl.65/>

Robustness

RoTBench: A Multi-Level Benchmark for Evaluating the Robustness of Large Language Models in Tool Learning

Authors: Junjie Ye et al. | **Year:** 2024

Why relevant: Evaluates robustness of tool selection, parameter identification and content filling under noise.

Official source: <https://aclanthology.org/2024.emnlp-main.19/>

Calibration

Uncertainty Calibration for Tool-Using Language Agents

Authors: Hao Liu et al. | **Year:** 2024

Why relevant: Studies uncertainty and calibration in tool-use decisions and execution paths.

Official source: <https://aclanthology.org/2024.findings-emnlp.978/>

Multi-agent

Learning to Use Tools via Cooperative and Interactive Agents

Authors: Zhengliang Shi et al. | **Year:** 2024

Why relevant: Separates tool selection, execution and action calibration among cooperating agents.

Official source: <https://aclanthology.org/2024.findings-emnlp.624/>

2. Verification checklist for every paper

- Open the official landing page.
- Match title and authors.
- Check year and venue.
- Check DOI/arXiv/ACL identifier.
- Confirm the link resolves to the same paper.
- Read the abstract/introduction enough to confirm relevance.
- Record the verified information in references.md.

3. Research synthesis

Taken together, the literature suggests that tool selection should not be evaluated as a single isolated click. It sits inside a pipeline involving instruction understanding, planning, retrieval, selection, parameter construction, execution and response generation. T-Eval explicitly decomposes tool utilization into sub-processes, while BFCL evaluates function calling across increasingly agentic settings. RoTBench adds robustness under noise, and ToolSandbox adds stateful and conversational dependencies. These perspectives provide a strong basis for an evaluation-focused repository.

4. Suggested categories in references.md

Survey & Review → Foundations → Tool Use & Function Calling → Tool Selection & Retrieval → Planning → Evaluation & Benchmarks → Robustness/Calibration → Recent Work.